# The Traffic Director

Load balancing and bottlenecks

SCAN / LEARN



## THE MAIN IDEA

A load balancer distributes incoming requests among healthy workers. It can reduce overload but cannot invent unlimited capacity.

## FOLLOW THE SYSTEM

**01 REQUESTS**

Many devices request the same service.

**02 BALANCER**

The front door selects a healthy worker.

**03 WORKERS**

Several servers complete the requests.

**04 FEEDBACK**

Health and load measurements change routing.

VISITORS → BALANCER → HEALTHY WORKERS

## COMMON MISCONCEPTION

Load balancing makes a slow database fast automatically.

## MORE ACCURATE

A shared bottleneck remains until its own capacity changes.

## THREE QUESTIONS TO KEEP

**PROBLEM:** How do we distribute incoming traffic?

**FAILURE:** One worker is flooded while others sit idle.

**IN REAL LIFE:** Shopping, video platforms and AI services.

## SOLO DISCOVERY

Route 12 numbered requests to 3 servers using round-robin.